

1. **Forwarding vs. Routing:** Explain the fundamental difference between *forwarding* and *routing*. Which action happens at nanosecond timescales in hardware, and which happens at longer timescales in software?

Forwarding

- **Definition:** The process of moving a packet from an incoming interface to the appropriate outgoing interface on a router or switch.
- **Scope:** It's a local action — the device already knows the correct output port from its forwarding table.
- **Timescale:** Happens at **nanosecond timescales in hardware** (using ASICs or fast switching hardware).

Routing

- **Definition:** The process of determining *which path* packets should take through the network. It involves building and maintaining routing tables using protocols (like OSPF, BGP, RIP).
- **Scope:** It's a global action — routers exchange information to decide the best path.
- **Timescale:** Happens at **longer timescales in software** (milliseconds to seconds, depending on convergence).

2. **Router Architecture:** Describe the phenomenon of **Head-of-the-Line (HOL) blocking**. Does this occur at the input ports or output ports of a router?

HOL blocking occurs **at input ports** when a packet at the front of a queue waits for a busy output port, blocking packets behind it that could go elsewhere.

3. **Longest Prefix Matching:** A router has the following forwarding table. Determine the outgoing link interface for the destination address 11001000 00010111 00010110 10100001.

- 11001000 00010111 00010*** ***** → Link 0
- 11001000 00010111 00011000 ***** → Link 1
- 11001000 00010111 00011*** ***** → Link 2
- *Otherwise* → Link 3

Destination: 11001000 00010111 00010110 10100001

Matches:

- 00010*** → Link 0 (5 bits match)
- 00011000 → Link 1 (8 bits match)

- 00011*** → Link 2 (5 bits match)

Longest match = Link 1

4. **Fragmentation:** Why does IPv6 *not* support fragmentation at intermediate routers, whereas IPv4 does? What is the specific benefit of removing this feature?

IPv6 removes router fragmentation to:

- Simplify router processing
- Increase forwarding speed

Only the sender fragments. **Benefit: faster routers, simpler hardware.**

5. **NAT (Network Address Translation):** Explain the specific problem that NAT was designed to solve regarding IP address exhaustion. How does a NAT router use port numbers to map multiple internal private IP addresses to a single external public IP address?

NAT solves IPv4 exhaustion (NAT allows multiple devices inside a private network to share a single public IP when accessing the internet, conserving scarce public addresses.) by mapping:

(Private IP, Port) ↔ (Public IP, Unique Port)

Example:

192.168.1.2:5000 → 73.5.6.7:30001
192.168.1.3:5000 → 73.5.6.7:30002

6. **IPv6 Header Format:** Compare the IPv4 and IPv6 headers. Name two fields that were *removed* in the IPv6 header to streamline processing, and explain why the "Header Checksum" field was deemed unnecessary in IPv6.

Removed fields:

- Header Checksum
- Fragmentation fields

Checksum unnecessary because link layer + TCP/UDP already do error checking.

7. **Link State vs. Distance Vector:** In a Link State algorithm (like Dijkstra's), what specific information is shared and with whom? Contrast this with a Distance Vector algorithm (like Bellman-Ford).

- **Link State:** Each router floods **link costs to all routers.**
- **Distance Vector:** Routers share **distance vectors only with neighbors.**

8. **Count-to-Infinity:** Explain the "Count-to-Infinity" problem in Distance Vector routing. Why does "bad news travel slow" when a link cost increases drastically?

When a link cost increases, routers slowly increment distance estimates because they rely on neighbors' outdated info. Hence **bad news travels slow**.

9. **BGP Path Selection:** A BGP router learns multiple routes to a destination. Order the following criteria from first to last in the selection process: (a) Shortest AS-PATH, (b) Local Preference, (c) Closest NEXT-HOP router.

BGP Path Selection Order

- 1) Local Preference
- 2) Shortest AS-PATH
- 3) Closest NEXT-HOP

10. **ICMP:** When you run a traceroute command, the source sends UDP segments with increasing TTL values. What specific ICMP error message does the *intermediate* router send back when the TTL expires?

ICMP Time Exceeded

11. **OpenFlow & SDN:** In an OpenFlow switch, forwarding is based on a "Match-Action" table. Give an example of a header field that can be matched (e.g., ingress port, dest IP) and an example of an action (e.g., drop, forward to port). How does this differ from a traditional destination-based forwarding table?

Match: Destination IP

Action: Forward to Port 3

Unlike traditional routing, it's **not only destination-based**.

12. **MAC vs. IP Addresses:** How does a MAC address differ from an IP address in terms of portability? If you move your laptop from a university network to a coffee shop, does your MAC address change? Does your IP address change?

- MAC: **Does not change**
- IP: **Changes** when moving networks

13. **ARP:** Host A wants to send an IP datagram to Host B on the same LAN but only knows B's IP address. Describe the steps Host A takes to resolve B's MAC address. What is the destination MAC address of the ARP query frame?

- A broadcasts ARP request
- B replies with MAC

Destination MAC of ARP query: **FF:FF:FF:FF:FF:FF**

14. **Switch Self-Learning:** Explain how a switch uses "Self-Learning" to populate its switch table. How does this reduce collision domains compared to a hub?

- Switch records **source MAC and ingress port** from frames.
- Each port becomes its own **collision domain**, unlike a hub.

15. **CSMA/CD:** In Ethernet's CSMA/CD protocol, what does a node do immediately after detecting a collision? What is the purpose of the "jam signal"?

- Send **jam signal**
- Stop transmitting
- Wait random backoff time

Jam signal ensures all nodes detect the collision.

16. **Link Layer Services:** The Link Layer can provide services like "Error Detection" and "Reliable Data Transfer." Why might a wireless link (like WiFi) require both of these services, while a wired fiber optic link might only use error detection and skip reliability?

- Wireless: noisy → needs **error detection + reliability**
- Fiber: very clean → **error detection only**

17. **VLANs:** A switch is configured with two VLANs (Engineering and Sales). How does a **trunk port** handle traffic between switches for different VLANs (specifically regarding the 802.1Q standard)?

Trunk adds a **VLAN tag** to frames so multiple VLANs travel over one link.

18. **Data Center Topology:** Why do modern data centers use specialized topologies (like Leaf-Spine) instead of a simple hierarchy? How does this help servers communicate with each other faster?

Provides **many equal-cost paths** → faster server-to-server communication, no bottlenecks.

19. **Load Balancing:** What is the main purpose of a **Load Balancer** in a data center? When you access a large website, why does your request go to a Load Balancer first instead of directly to a web server?

Distributes traffic across servers for:

- Reliability
- Scalability

20. **Redundancy (Multiple Paths):** Why is it important to have multiple physical links (paths) between switches in a data center? How does this help if a link fails or if there is heavy traffic?

Provides:

- Fault tolerance
- Load sharing during congestion

21. **Hidden Terminal Problem:** Explain how two wireless nodes (A and C) can both communicate with a central node (B) but interfere with each other because they cannot detect each other's transmissions.

A and C can't hear each other but both send to B → collisions at B.

22. **CSMA/CA:** Why does 802.11 (WiFi) use **Collision Avoidance (CA)** instead of **Collision Detection (CD)** like Ethernet?

WiFi can't detect collisions while transmitting → must **avoid** them.

23. **RTS/CTS:** Describe the **RTS/CTS** (Request to Send / Clear to Send) exchange. How does this mechanism solve the hidden terminal problem?

A sends RTS → B replies CTS → others stay silent → prevents hidden terminal collisions.

24. **TCP over Wireless:** Why does TCP often perform poorly in wireless networks, even when there is no congestion? How does it misinterpret packet loss caused by fading or bit-errors?

TCP thinks loss = congestion. Wireless loss is from fading/errors → unnecessary slowdown.

25. **Wireless Signal Propagation:** Describe two physical impairments that make wireless transmission more difficult than wired transmission (e.g., Multipath Propagation, Path Loss/Fading). How does "Multipath" cause a signal to be received multiple times at the destination?

- Path loss/fading
- Multipath

Multipath: signal reflects → arrives multiple times with delays.

26. **4G/5G Architecture:** What are the specific roles of the **Home Subscriber Service (HSS)** and the **MME (Mobility Management Entity)** in a cellular network?

- **HSS:** User database, authentication

- **MME:** Mobility, session management

27. **Mobile IP:** Describe the difference between Indirect Routing (via a Home Agent) and Direct Routing. Which one suffers from the "triangle routing" problem?

- Indirect routing via Home Agent → **triangle routing problem**
- Direct routing avoids it

28. **Subnet Calculation:** Given the IP block 192.168.10.0/24, design a subnetting scheme to support four departments: Admin (50 hosts), Sales (40 hosts), IT (20 hosts), and Guest (60 hosts). Provide the specific subnet masks and network addresses for each.

Dept	Hosts	Subnet	Mask
Guest	60	192.168.10.0/26	255.255.255.192
Admin	50	192.168.10.64/26	255.255.255.192
Sales	40	192.168.10.128/26	255.255.255.192
IT	20	192.168.10.192/27	255.255.255.224

29. **Middlebox Architecture:** Middleboxes (like NAT) perform functions that modify packet headers in transit. How does this violate the traditional **End-to-End Principle** of the Internet? Why is NAT still strictly necessary in IPv4 networks despite this violation?

NAT modifies headers → breaks end-to-end transparency. Still necessary due to IPv4 shortage.

30. **IPv6 Adoption:** For a large-scale IoT (Internet of Things) deployment requiring thousands of sensors, why is IPv6 preferable to IPv4 regarding header processing and address availability?

- Huge address space
- Simpler header → faster processing

31. **VLAN Design:** You have a 3-story building with users from two different departments on every floor. Explain how **VLANs** allow you to logically separate these departments using only one switch per floor, rather than physically buying separate switches for each department.

Logically separates departments on same physical switch using VLAN IDs.

32. **Collision Domains:** If you upgrade a legacy network from Hubs to Layer 2 Switches, how does this affect the number of Collision Domains and Broadcast Domains? Why does this improve bandwidth?

- Collision domains: **increase (good)**
- Broadcast domains: unchanged
- More bandwidth per device

33. **SDN Architecture:** In a Software-Defined Network, the control plane is physically separate from the data plane. Describe the role of the **Southbound API** (e.g., OpenFlow) versus the **Northbound API** (connecting to applications).

- Southbound (OpenFlow): Controller → switches
- Northbound: Apps → controller

34. **Routing Protocol Selection:** Compare **OSPF** and **BGP**. Which protocol is appropriate for internal campus routing (Intra-AS) and which is appropriate for routing between ISPs (Inter-AS)? Justify your choice based on the metrics they use (Link State vs Policy/Path).

- **OSPF:** Intra-AS (link state)
- **BGP:** Inter-AS (policy/path based)

35. **SDN Utility:** In a data center environment that requires frequent, automated changes to traffic flows, why is a **Software-Defined Networking (SDN)** architecture often preferred over a traditional distributed control plane?

Centralized control allows fast, automated traffic changes.

36. **WiFi Capacity:** In a high-density environment (like a lecture hall with 400 students), why is a single Wireless Access Point insufficient even if it has the signal range to cover the room? (Discuss the shared medium nature of WiFi). How do multiple APs on non-overlapping channels (1, 6, 11) solve this?

WiFi is shared medium. Multiple APs on channels 1/6/11 create parallel capacity.

37. **Wireless Technology Selection:** You are designing a "Smart Office" system. You need to connect low-power door sensors that run on coin-cell batteries. Would you choose **WiFi (802.11)** or **Bluetooth (802.15)**? Justify your choice based on power consumption and typical range.

Bluetooth (802.15) — very low power, ideal for coin-cell sensors.